



S E R V E W E L L C O .

Loan Management System Independent Assessment

A technical and product review of the Jasiri Lending Software platform

PREPARED FOR



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IT Consultant · Serve Well Co.

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CONFIDENTIAL — prepared solely for Mular Credit. Contains security-sensitive findings; handle accordingly.

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1 Executive summary

Jasiri is a capable lending platform with a genuinely strong analytical core and a well-shaped product model. This review — conducted live on the production tenant — finds it held back by three things: onboarding friction that inflates the time from application to disbursement; a handful of unfinished areas presented as finished; and, most urgently, security exposures on the live system that warrant immediate attention. None of these is fatal. All are fixable, and the same fixes are the roadmap to a fast, identity-first origination engine.

THE HEADLINE

Mular's greatest opportunity is turnaround. Two capabilities collapse the time from application to disbursement without loosening control: **IPRS-first identity** (an ID number returns verified particulars — the very thing that impressed the CEO when he handed over only his own ID) and **internal statement scoring that assigns a starting limit and matches the customer to the right product — with reason codes**. Together they move expensive human judgement to the few cases that need it, instead of every case.

Findings at a glance

16 findings across 11 module groups and 60+ screens: **1 Critical** · **6 High** · **7 Medium** · **2 Low**

Module maturity

Module group	Standing	In a line
Dashboards & Analytics	Strong	The platform's best asset — broad, well-organised analytics.
Leads	Strong	Lead management, aging, converted and insight views all present.
Audit & Logs	Adequate	Data-change and user-activity logs exist — a good foundation.
Loan Products & Credit	Adequate	Sensible tiered products; penalty configuration is broken (F-08).
Loaning & Approvals	Adequate	Workable, but gated behind a long manual approval chain.
Reports & Finance	Adequate	Content is right; presentation and access model are dated.
Registry & Onboarding	Needs work	Friction, weak validation, and a hard dead-lock (F-01...F-06).
Administration & Security	At risk	Privilege self-escalation and non-expiring sessions (S-01, S-02).
System Settings	At risk	Placeholder — every item redirects to the dashboard (S-03).

2 Scope and method

What was reviewed. The live Jasiri Lending Software tenant operated for Mular Credit, exercised end-to-end on 24 July 2026 by Emmanuel Birgen as a provisioned Administrator. The review covered all eleven module groups — Dashboard, Leads, Registry, Loaning, Credit Settings, Accounts & Finance, Reports, Drafts, Administration, System Settings, and Audit & Logs — across more than sixty screens, working the real customer-onboarding and loan journeys rather than inspecting screens in isolation.

How severity is rated. Each finding is rated by business impact and likelihood: **Critical** (control-defeating, act now), **High** (blocks users or exposes the business), **Medium** (friction or quality), and **Low** (polish).

On the security findings. Section 8 was found and confirmed in the course of normal testing and is disclosed here responsibly and privately. No customer data was exfiltrated or altered; the only action taken was the minimum needed to demonstrate that a role could be self-assigned. We recommend the items in Section 8 be treated as time-sensitive.

3 What Jasiri does well

A fair critique begins with credit where it is due — and Jasiri earns it in several places.

3.1 A comprehensive analytics core

The Analytics Dashboard is the platform's strongest asset. Beyond Regional and Branch performance, it carries Loan Product Overview, Customer Retention, Repayment Trends, Customer Age Distribution, Business Sector Distribution, Guarantor Age Analysis, Payer Type Analysis, Customer Distribution by County, Marital Status Breakdown, and NPL Analysis. That is a genuinely complete analytical surface, and it sets the bar to meet — and then exceed — with a more cinematic, presentation-grade analysis studio.

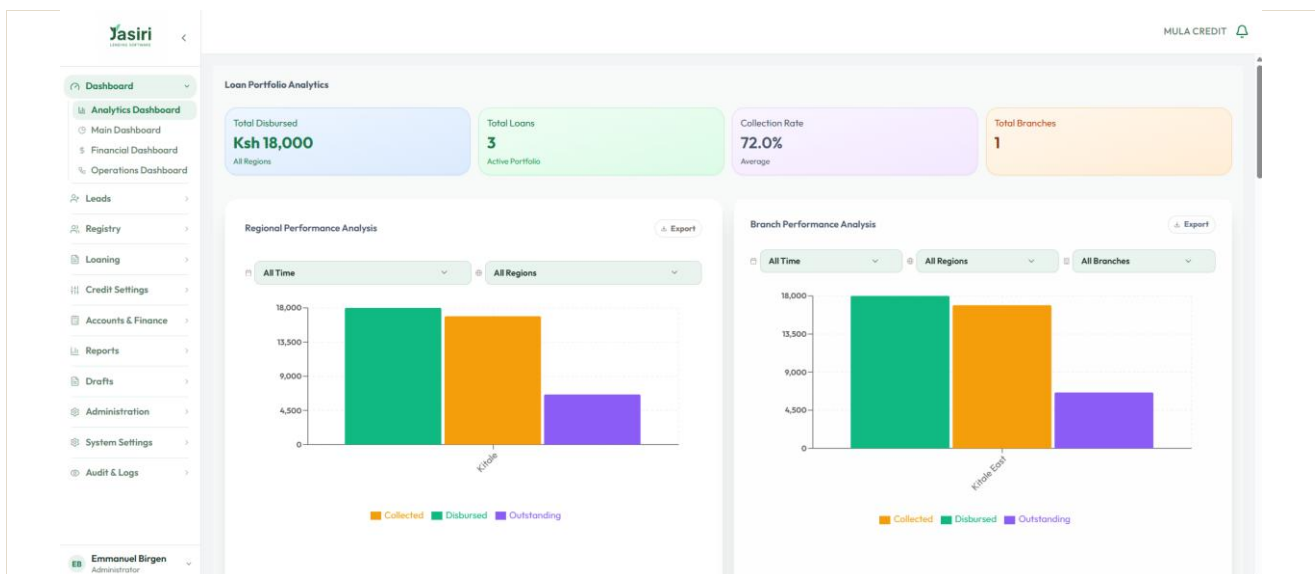


Figure 1 — The Analytics Dashboard: broad, well-organised, and the platform's clear high point.

3.2 Input validation done right, in the right places

- **Business age.** A business under six months old is rejected at the business-info step — exactly the guardrail a micro-lender wants at the door.
- **Phone formats.** Both the primary and alternative mobile numbers are format-checked ("10–15 digits"), catching a common data-quality problem early.

3.3 Institutional memory of every customer touch

The customer record carries interactions and dispositions saved by call-centre agents, alongside approval history, limit-adjustment history, and scored-amount history. This is a real strength and worth replicating well — it is the institution's memory of every conversation and decision about a customer.

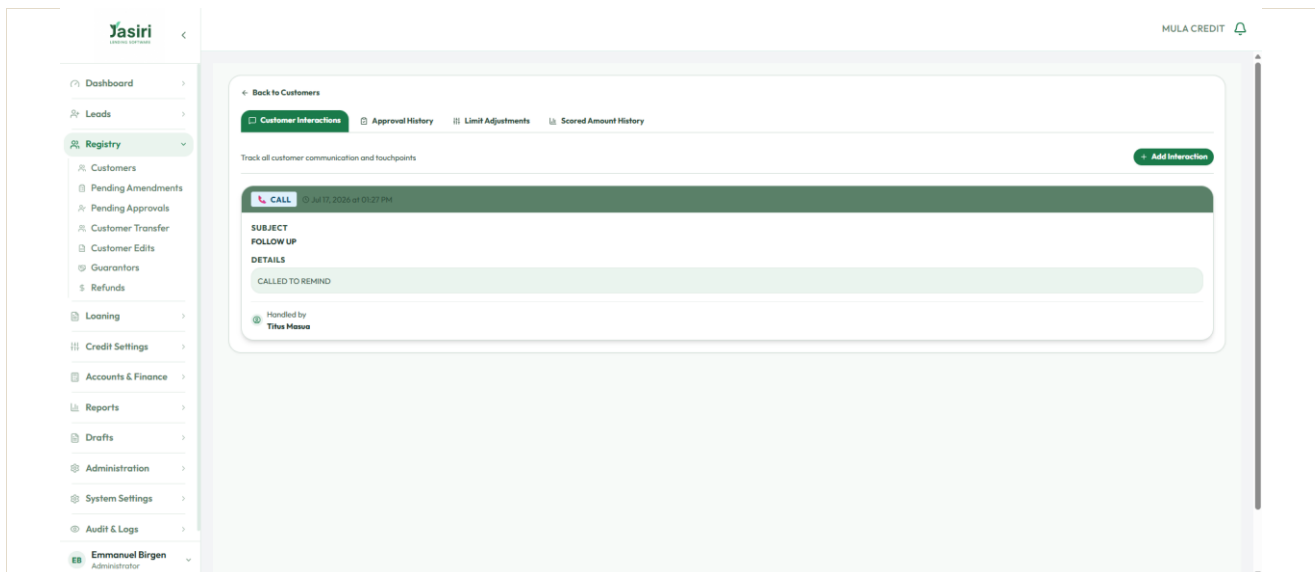


Figure 2 — Customer interactions, approval history, limit adjustments, and scored-amount history on the customer record.

3.4 A workable approval workflow, and a real finance module

- **Configurable workflow statuses** and approval messages run across the loan chain — a good catch and a sound idea.
- **A fuller finance surface than many peers ship:** chart of accounts, journal vouchers, bank and manual reconciliation, a financial-transactions ledger, and promise-to-pay reports.
- **Report access codes** show the right security instinct — reports should be gated (we refine how, in F-10).

4 The core opportunity — origination and turnaround

Mular's operating reality is a long runway from lead to disbursement: leads are captured, converted, and then passed through multiple approval stages — much of it before a loan application even exists. It is thorough, and it is slow. The single largest opportunity is to collapse that turnaround without losing an ounce of control. Two capabilities do most of the work.

① IPRS-FIRST IDENTITY

Instead of hand-keying a customer's particulars and validating them after the fact, the officer enters an ID number and the system returns the verified names and details — the same principle that let the CEO hand over only his ID number and watch his own record return. Applied to the **borrower, the guarantor, and the next-of-kin**, it removes the single largest source of both friction and fabricated data at once — and closes F-03 and F-04 by construction.

② INTERNAL SCORE → STARTING LIMIT → PRODUCT MATCH

Rather than pushing every applicant through the full manual chain to discover their standing, an M-Pesa statement is analysed into an **internal score**; from that score the customer is given a **starting loan limit** and matched to the product bucket and term that fit — FADHILI, KUZA, or INUKA — each recommendation carrying its **reason codes** (why this limit, why this product). The officer sees, on one screen, exactly what the customer qualifies for and why, before the approval chain begins.

Together, these move the expensive human judgement to the end — on the handful of cases that genuinely need it — instead of the front, on every case. That is how the time from application to disbursement falls, and it is exactly where Serve Well Co. and the BirgenAI engine suite are built to help.

5 Registry and onboarding

Onboarding is where a lending platform earns or loses its operators, and it is where Jasiri needs the most work. The journey is functional but brittle: validation fires too late, drafts are lost, and one failure surfaces the raw database. Each finding below is small on its own; together they inflate turnaround and, in one case, strand the user entirely.

Figure 3 — The new-customer business-info step, where several of the findings below first appear.

F-01 Save-as-draft fails and loses earlier steps

HIGH

WHAT HAPPENS

On the multi-step new-customer form, “Save as draft” at Stage 2 returns “Failed to save draft.” After a refresh, every entry — including the completed Stage 1 — is gone.

IMPACT

An officer who is interrupted loses all captured data and must re-key from scratch. In the field, on intermittent mobile data, this all but guarantees abandoned registrations and inflated turnaround. It also implies steps are not persisted server-side as they are completed.

RECOMMENDATION

Persist each step to a draft record the moment it is completed (autosave), so a draft is always recoverable and the form resumes where it stopped. Never advance a wizard whose previous step was not saved.

F-06 A raw database error is shown to the user — and dead-locks them**HIGH****WHAT HAPPENS**

After every step was completed correctly, submission failed with a raw database message: "null value in column "branch_id" of relation ... violates not-null constraint." The screen showed the database's words, with no plain-language reason and no way forward.

IMPACT

A real officer dead-locks precisely here: everything looks complete, yet an internal constraint (no branch attached) blocks the submission with no guidance. This single class of error strands users, generates support calls, and leaks internal schema details to the outside.

RECOMMENDATION

Never surface raw exceptions. Catch them, translate to plain language ("This application isn't linked to a branch — choose the customer's branch to continue"), and point to the fix. An embedded assistant should let a stuck user simply ask "why did this fail?" and get a human answer.

F-03 Next-of-kin accepts anything — even the borrower's own details**HIGH****WHAT HAPPENS**

The next-of-kin step performs no validation. The borrower's own name and phone number were entered as the next-of-kin and the form advanced without objection.

IMPACT

Next-of-kin is a recovery and tracing safeguard. If it can equal the borrower, it is worthless, and the customer record is polluted at the source. The same absence of checks would let guarantor details be fabricated.

RECOMMENDATION

Validate that next-of-kin differs from the borrower (name, phone, ID) and require a stated relationship. Better still, **capture next-of-kin and guarantor by ID and auto-populate verified identity** — removing the very opportunity to invent the data.

F-02 Town is asked before County, and starts empty**MEDIUM****WHAT HAPPENS**

The Town / City selector sits to the left of County and is empty, because its options depend on a County that has not yet been chosen. Users click Town first, find nothing, and stall.

IMPACT

Predictable confusion at the very first data-entry screen — the worst place to lose the operator's confidence.

RECOMMENDATION

Order County first; reveal or enable Town only once a County is set, and populate it from that county. A control that is disabled-until-ready reads far more clearly than one that is simply empty.

F-04 Date of birth is not enforced until the final submit**MEDIUM****WHAT HAPPENS**

The wizard allows the operator to proceed past the identity step without a date of birth, then rejects the final submission with "Please fix missing fields (first: dateOfBirth)."

IMPACT

The user is told at the end about a field they passed at the beginning — the least helpful moment possible. Combined with the draft-loss issue (F-01), it can cost the entire entry.

RECOMMENDATION

Enforce mandatory fields at their own step and block “Next” until they are satisfied. Fail early and in context, never at the finish line.

F-05 Verification photos are silently mandatory**MEDIUM****WHAT HAPPENS**

Officer / client verification photos (first officer + client, second officer + client, both officers + client) are required, but the requirement only appears as a submit-time error after everything else is complete.

IMPACT

The same late-failure pattern, and it is unclear whether the requirement is configurable per lender or hard-coded for all.

RECOMMENDATION

Surface the requirement at the photo step itself, and make each photo’s necessity a visible, per-lender configurable rule.

F-13 Transfer screens have no way back**MEDIUM****WHAT HAPPENS**

In Customer Transfers, “Create Transfer” opens the transfer-details page, but “Cancel” does not return to the previous list. “Review” opens the “Execute Transfer Now” page with no back button at all — the only way out is to re-click the sidebar.

IMPACT

Broken spatial navigation: users lose their place and their trust in the flow, and re-entering from the sidebar risks losing context.

RECOMMENDATION

Give every secondary screen an explicit Back / Cancel that returns to the originating list with its state intact.

F-07 Business-location map is slow to appear**LOW****WHAT HAPPENS**

On the Business Location step, the map (and two dependent required fields) takes a long time to render.

IMPACT

Perceived slowness at a required step; officers may wait unnecessarily or mis-enter while it loads.

RECOMMENDATION

Lazy-load the map behind a lightweight placeholder and capture coordinates with a fast “use current location” that does not block on a full map render.

SERVE WELL RECOMMENDS

An **ID-first onboarding flow** with step-level validation, true autosave drafts, and plain-language errors backed by an embedded assistant. Most of the findings above disappear not by patching each one, but by changing the shape of the journey: verify identity up front, save as you go, and never let a user reach a wall they cannot read.

6 Loan products and credit settings

Mular's catalogue is well-conceived: three tiers by ticket size — INUKA (KES 1,000–5,000), KUZA (6,000–10,000), and FADHILI (11,000 and above) — each with weekly terms from four to eight weeks, interest scaling with tenor (25% at four weeks up to 50% at eight), and processing charged either as a percentage (FADHILI, 5%) or a flat fee (KES 500 on INUKA and KUZA). A global joining fee of KES 300 is applied once per new customer. The tiering is exactly the right shape for a graduating micro-borrower.

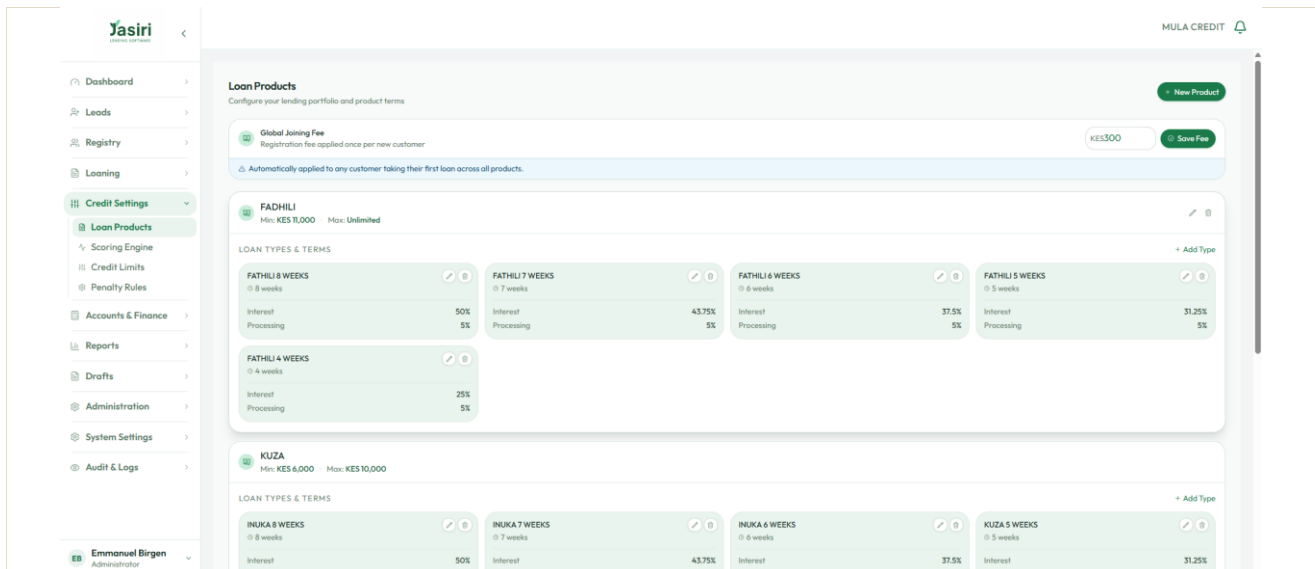


Figure 4 — The FADHILI / KUZA / INUKA product tiers, with per-term interest and processing charges.

F-08 Penalty Rules settings fail to load

HIGH

WHAT HAPPENS

Opening Credit Settings → Penalty Rules shows “Error Loading Penalty Settings / Failed to load penalty settings / Retry Connection.” The page does not recover.

IMPACT

Penalties are core credit configuration. A lender cannot review or change how arrears are charged. For a micro-lender whose margin depends on repayment discipline, an inaccessible penalty engine is material.

RECOMMENDATION

Fix the endpoint, and treat the availability of credit configuration as a monitored health check — not something a user discovers is broken.

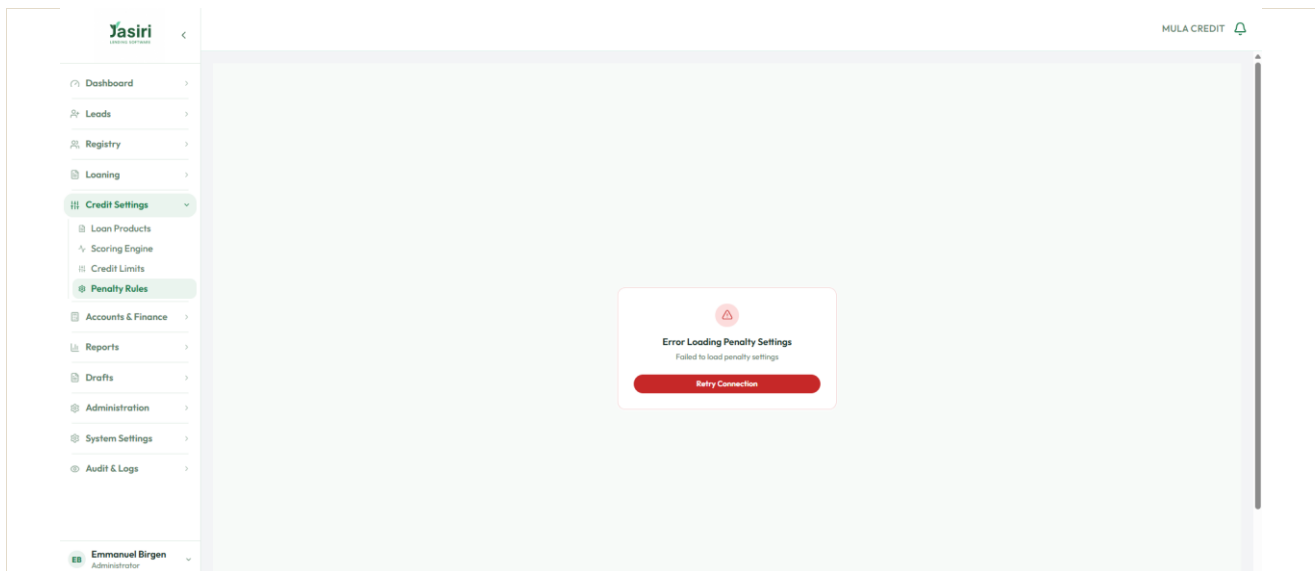


Figure 5 — Credit Settings → Penalty Rules fails to load.

SERVE WELL RECOMMENDS — SEPARATE CHARGES FROM PRODUCTS

In the strongest design, charges are decoupled from products. The **joining fee belongs to borrower-level rules** (alongside eligibility such as age limits), and **processing fees attach to each product as distinct, individually-priced charges** — so pricing flexes per product and per lender without ever editing the product itself. This is also what makes reason-coded, limit-driven product matching (Section 4) clean to implement.

7 Reports and finance

F-09 Income statement is sound but visually dated

MEDIUM

WHAT HAPPENS

The income statement is numerically correct — revenue by product, processing and joining fees, penalties, and totals — but the presentation is a plain generic table with no branding, no visual hierarchy, and no period framing beyond a header line.

IMPACT

This is a document a lender shows owners, auditors, and funding partners. Its appearance is the institution's brand, and "correct but plain" undersells a serious business.

RECOMMENDATION

A branded, properly typeset statement in Mular's identity — period comparatives, clear sub-totals, share-of-revenue — exporting cleanly to PDF and Excel.

Item	July	YTD
REVENUE		
INTEREST ON INUKA 4 WEEKS	KES 0.00	KES 1,250.00
Total inuka 4 weeks	KES 0.00	KES 1,250.00
INTEREST ON INUKA 6 WEEKS	KES 0.00	KES 750.00
Total inuka 6 weeks	KES 0.00	KES 750.00
INTEREST ON INUKA 5 WEEKS	KES 30.00	KES 1,905.00
Total inuka 5 weeks	KES 30.00	KES 1,905.00
LOAN PROCESSING FEE	KES 0.00	KES 1,500.00
Total loan processing fee	KES 0.00	KES 1,500.00
JOINING FEE	KES 0.00	KES 900.00
Total joining fee	KES 0.00	KES 900.00
PENALTIES	KES 0.00	KES 0.00
Total penalties	KES 0.00	KES 0.00
TOTAL REVENUE	KES 30.00	KES 6,305.00

Figure 6 — The income statement: numerically sound, visually plain.

F-10 Report access is by shared code, not by user**MEDIUM****WHAT HAPPENS**

Reports are gated behind access codes — a good security instinct — but access is granted by code rather than by identity, and is not scoped per report per user.

IMPACT

Codes are shareable and coarse. They do not answer the question a regulator or owner will ask: who, specifically, can open which report?

RECOMMENDATION

A Report Access Manager that binds report visibility to the user at creation — choose exactly which reports each user may open and how they are scoped (branch, region, own-book). Auditable and revocable.

Figure 7 — The current report-access approach.

F-11 Manual reconciliation is not clearly guided**MEDIUM**

WHAT HAPPENS

Manual reconciliation exists, but the path — see an unmatched payment, understand where it came from, then attach it to the right customer — is not clearly led.

IMPACT

Reconciliation mistakes are money mistakes, and an unclear flow makes them more likely and harder to trace.

RECOMMENDATION

A clear three-beat flow — show the money and its source → search the customer → reconcile to that account — with a visible audit trail on every match.

8 Security findings — responsible disclosure

These findings were confirmed during ordinary testing and are shared privately so they can be fixed quickly. They are the most important items in this document. We recommend they be treated as time-sensitive, and we would be glad to help remediate them.

S-01 An Administrator can promote itself to Super Admin

CRITICAL
WHAT HAPPENS

An account provisioned only as Administrator was able to assign itself the Super Admin role from the role dropdown and obtain full access. The reviewer's own account — provisioned as Administrator (see the account badge) — now appears in the "Superadmin · Full access" group alongside the CEO.

IMPACT

The most serious finding in this review. Any admin-tier user can silently seize total control of the tenant — customer data, credit settings, finance, and other users, including the CEO's own account. It defeats every other control in the system.

RECOMMENDATION

Authorise role changes server-side against the actor's own authority (a user may never grant a role above their own); remove Super Admin from the assignable options in both UI and API; require four-eyes approval for any elevation; and log and alert on every role change. This was tested responsibly — no data was accessed or altered beyond demonstrating the assignment.

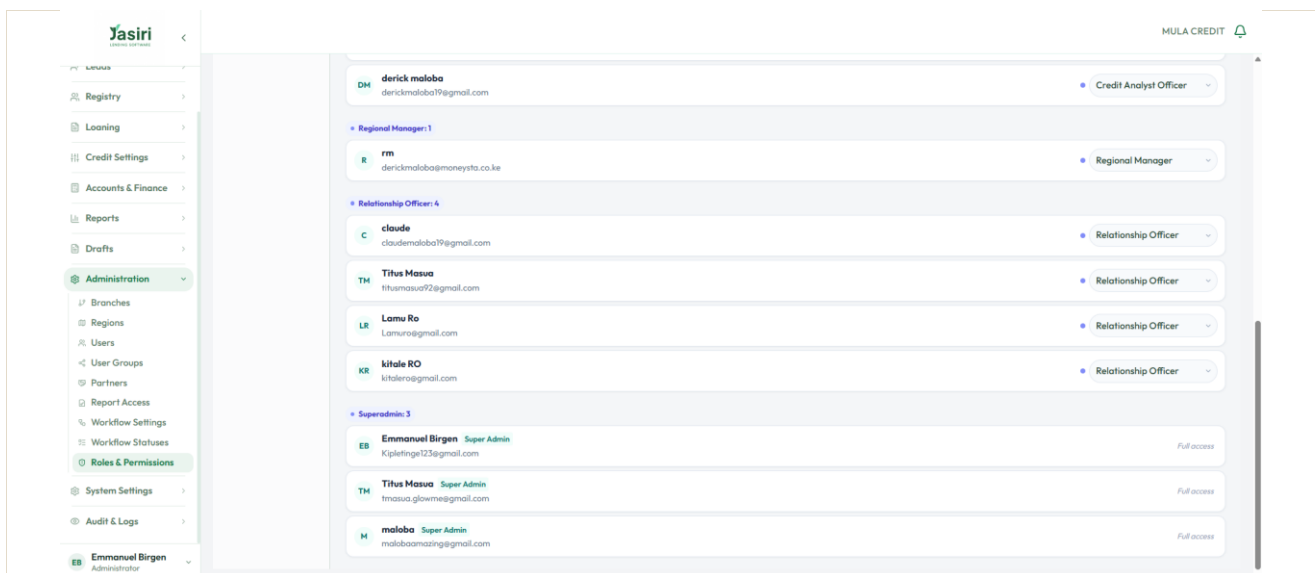


Figure 8 — The reviewer's account, provisioned as Administrator, now listed under "Superadmin · Full access." (Staff details can be cropped before this report is shared onward.)

S-02 Sessions never expire

HIGH

WHAT HAPPENS

Sessions do not time out. The reviewer remained logged in for roughly two days across closing the laptop and the browser, without ever being asked to re-authenticate.

IMPACT

A shared or stolen device stays authenticated indefinitely, and no sensitive action asks for re-authentication. Combined with S-01, a walk-up attacker inherits full control.

RECOMMENDATION

Introduce an idle timeout and an absolute session lifetime; require re-authentication (or a step-up one-time code) for sensitive actions such as role changes, disbursement, and settings; and consider daily reusable one-time codes for staff sign-in.

S-03 System Settings are placeholders presented as live

HIGH

WHAT HAPPENS

The entire System Settings section — General, Email, SMS, and the rest — is non-functional: every item redirects to the dashboard.

IMPACT

A live product is presenting configuration surfaces that do not exist. An operator may believe they have configured something (an SMS sender, an email template) that was never applied — a silent-failure risk with real operational consequences.

RECOMMENDATION

Either build these settings or clearly mark them "coming soon" and hide the dead links. Never present a control that does nothing.

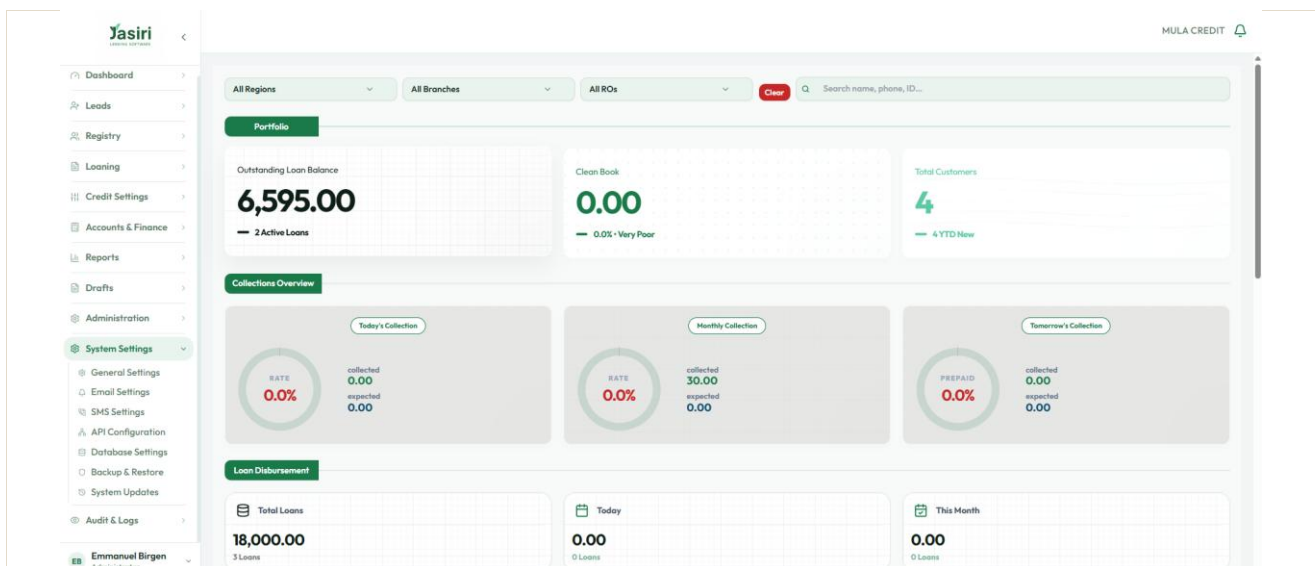


Figure 9 — Every System Settings item redirects to the dashboard.

9 Integration and single sign-on

A specific request from Mular's leadership: the platform should work seamlessly alongside separate systems — a HR system, an accounting system, a call-centre system — each with its own front door, but with the user authenticated once. This is a single-sign-on and identity problem, and it is very solvable.

- **One identity layer.** A user signs in once and that identity is honoured across every connected system. Each system may keep its own login page as an entry point, but all defer to the shared session — no repeated logins, no divergent user lists.
- **Engines as APIs.** The differentiating capabilities — IPRS identity, statement analysis, credit scoring, and M-Pesa Ratiba auto-repayments — are exposed as clean APIs, so they can be embedded into whichever system needs them (including systems Mular already runs) rather than trapped inside one screen.

This is the point at which a platform stops being “an LMS” and becomes the spine the rest of the business plugs into — and it is precisely the shape Serve Well Co. and the BirgenAI engines are built to deliver.

10 Recommended priorities

A pragmatic order of work — stabilise first, then differentiate.

Horizon	Focus	Items
NOW	Stabilise & secure	Lock down role self-escalation and session expiry (S-01, S-02); stop showing raw DB errors and fix branch attachment (F-06); restore penalty configuration (F-08); make drafts persist (F-01).
NEXT	Collapse turnaround	IPRS-first onboarding for borrower, guarantor and next-of-kin; internal-score → starting limit → product match with reason codes; step-level validation (F-02, F-04, F-05); transfer navigation (F-13).
LATER	Differentiate & polish	A cinematic Analysis Studio; branded income statement and a per-user Report Access Manager (F-09, F-10); guided reconciliation (F-11); unified SSO and engine APIs; step-up one-time codes; retire placeholder settings (S-03).

11 In closing

Jasiri is a capable platform with a strong analytical core and a sensible product model. It is held back by onboarding friction that inflates turnaround, by a few unfinished areas presented as finished, and — most urgently — by security exposures on the live system that need prompt attention. None of this is fatal; all of it is fixable, and the fixes double as the roadmap to a genuinely fast, identity-first origination engine.

Serve Well Co., with the BirgenAI engine suite, is positioned to deliver exactly that: identity-first onboarding, score-driven limits and reason-coded product matching, M-Pesa auto-repayment rails, and the integration spine that lets Mular's systems act as one. We would welcome the opportunity to take the priorities in Section 10 forward with you.

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A Appendix — findings register

ID	Severity	Finding
S-01	CRITICAL	An Administrator can promote itself to Super Admin
S-02	HIGH	Sessions never expire
S-03	HIGH	System Settings are placeholders presented as live
F-01	HIGH	Save-as-draft fails and loses earlier steps
F-06	HIGH	A raw database error is shown to the user — and dead-locks them
F-03	HIGH	Next-of-kin accepts anything — even the borrower's own details
F-08	HIGH	Penalty Rules settings fail to load
F-13	MEDIUM	Transfer screens have no way back
F-02	MEDIUM	Town is asked before County, and starts empty
F-04	MEDIUM	Date of birth is not enforced until the final submit
F-05	MEDIUM	Verification photos are silently mandatory
F-09	MEDIUM	Income statement is sound but visually dated
F-10	MEDIUM	Report access is by shared code, not by user
F-11	MEDIUM	Manual reconciliation is not clearly guided
F-07	LOW	Business-location map is slow to appear
F-12	LOW	Scroll position carries across module switches (minor)

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